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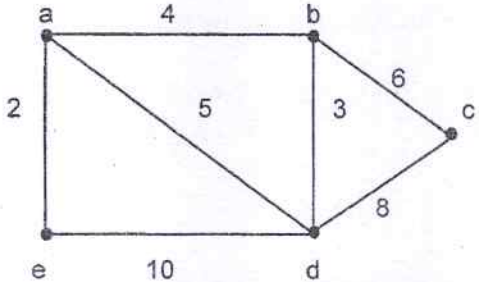
## ODD END SEMESTER EXAMINATION DECEMBER 2024

Name of the Course: **B.Tech.**Semester: **Third**Name of the Paper: **Discrete Structures and Combinatorics**Paper Code: **TMA 316**Time: **3 Hours**Maximum Marks: **100**

Note:

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks.

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |            |      |      |      |       |        |          |   |   |   |   |          |   |     |      |      |      |       |        |          |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------|------|------|-------|--------|----------|---|---|---|---|----------|---|-----|------|------|------|-------|--------|----------|
| <b>Q1</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | (20 marks) |      |      |      |       |        |          |   |   |   |   |          |   |     |      |      |      |       |        |          |
| (a)       | Let $N = \{1, 2, 3, \dots\}$ and a Relation is defined in $N \times N$ as follows: $(a, b)$ is related to $(c, d)$ iff $ad = bc$ , then show whether $R$ is an equivalence relation or not.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |            | CO1  |      |      |       |        |          |   |   |   |   |          |   |     |      |      |      |       |        |          |
| (b)       | Let $A = B = C = R$ and let $f: A \rightarrow B$ and $g: B \rightarrow C$ be defined by $f(a) = a+3$ and $g(b) = 2b+1$ . Find<br>(i) $\text{gof}(5)$ (ii) $\text{fog}(5)$ (iii) $\text{gof}(-5)$ (iv) $\text{fog}(-5)$ (v) $(\text{gof})x$ (vi) $(\text{fog})x$                                                                                                                                                                                                                                                                                                                                                                                                       |            |      |      |      |       |        |          |   |   |   |   |          |   |     |      |      |      |       |        |          |
| (c)       | Draw the Hasse diagram of $D_{36}$ . Let the sets $A = \{9, 18\}$ and $B = \{4, 6, 9\}$ . Find, if they exist,<br>(i) Upper bounds of $A$ and $B$<br>(ii) Lower bounds of $A$ and $B$<br>(iii) Supremum (lub) of $A$ and $B$<br>(iv) Infimum (glb) of $A$ and $B$ .                                                                                                                                                                                                                                                                                                                                                                                                   |            |      |      |      |       |        |          |   |   |   |   |          |   |     |      |      |      |       |        |          |
| <b>Q2</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | (20 marks) |      |      |      |       |        |          |   |   |   |   |          |   |     |      |      |      |       |        |          |
| (a)       | A production company has three units A, B and C which produce 35%, 40% and 25% respectively of the production. Experience shows that 5% items produced by A are defective, 2% items produced by B and 3% items produced by C are defective. An item is selected at random and found to be defective. Find the probability that it was produced by B.                                                                                                                                                                                                                                                                                                                  |            | CO2  |      |      |       |        |          |   |   |   |   |          |   |     |      |      |      |       |        |          |
| (b)       | The random variable $X$ has the following probability function values <table border="1" style="margin-left: 20px;"> <tr> <td><math>x</math></td> <td>0</td> <td>1</td> <td>2</td> <td>3</td> <td>4</td> <td>5</td> <td>6</td> <td>7</td> </tr> <tr> <td><math>P(X=x)</math></td> <td>0</td> <td><math>b</math></td> <td><math>2b</math></td> <td><math>2b</math></td> <td><math>3b</math></td> <td><math>b^2</math></td> <td><math>2b^2</math></td> <td><math>7b^2+b</math></td> </tr> </table><br>(i) Find the value of $b$ .<br>(ii) Determine the distribution function of random variable $X$ .<br>(iii) Evaluate $P(X < 6)$ , $P(X \geq 6)$ and $P(0 < X < 5)$ . |            |      | $x$  | 0    | 1     | 2      | 3        | 4 | 5 | 6 | 7 | $P(X=x)$ | 0 | $b$ | $2b$ | $2b$ | $3b$ | $b^2$ | $2b^2$ | $7b^2+b$ |
| $x$       | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1          |      | 2    | 3    | 4     | 5      | 6        | 7 |   |   |   |          |   |     |      |      |      |       |        |          |
| $P(X=x)$  | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $b$        | $2b$ | $2b$ | $3b$ | $b^2$ | $2b^2$ | $7b^2+b$ |   |   |   |   |          |   |     |      |      |      |       |        |          |
| (c)       | If $X$ is a Poisson variate such that $P[X = 1] = 2P[X = 2]$<br>Find mean and variance. Also find $P[X = 0]$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |            |      |      |      |       |        |          |   |   |   |   |          |   |     |      |      |      |       |        |          |
| <b>Q3</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | (20 marks) |      |      |      |       |        |          |   |   |   |   |          |   |     |      |      |      |       |        |          |
| (a)       | (i) Write the symbolic representation and give its contrapositive, converse and the inverse statements of "If it rains today, then I buy an umbrella".<br>(ii) Show that $(p \rightarrow r) \wedge (q \rightarrow r)$ and $(p \vee q) \rightarrow r$ are logically equivalent.                                                                                                                                                                                                                                                                                                                                                                                        |            | CO3  |      |      |       |        |          |   |   |   |   |          |   |     |      |      |      |       |        |          |
| (b)       | Prove by mathematical induction $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |            |      |      |      |       |        |          |   |   |   |   |          |   |     |      |      |      |       |        |          |
| (c)       | Check the validity of the following argument:<br>"If it is Saturday today, then we play soccer or basketball. If the soccer field is occupied, we don't play soccer. It is Saturday today, and the soccer field is occupied. Therefore, we play basketball or volleyball."                                                                                                                                                                                                                                                                                                                                                                                            |            |      |      |      |       |        |          |   |   |   |   |          |   |     |      |      |      |       |        |          |
| <b>Q4</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | (20 marks) |      |      |      |       |        |          |   |   |   |   |          |   |     |      |      |      |       |        |          |
| (a)       | Prove that the set $\{0, 1, 2, 3, 4, 5\}$ is a finite abelian group under addition modulo 6. What will happen if the set is $\{1, 2, 3, 4, 5\}$ ?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            | CO4  |      |      |       |        |          |   |   |   |   |          |   |     |      |      |      |       |        |          |

|     |                                                                                                                                                                                                                                                                                                                                                                                                            |            |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| (b) | <p>In how many different orders can 3 men and 3 women be seated in a row of 6 seats if</p> <p>(i) Any one may sit in any of the seats.<br/> (ii) Men occupy the first three seats and women occupy the last three seats.<br/> (iii) The first and the last seats must be filled by men.<br/> (iv) All members of same sex are seated in adjacent seats.<br/> (v) Men and women are seated alternately.</p> | CO5        |
| (c) | <p>Solve the recurrence relation by Characteristic roots method.</p> $a_n - 7a_{n-1} + 12a_{n-2} = n \cdot 4^n$                                                                                                                                                                                                                                                                                            |            |
| Q5  |                                                                                                                                                                                                                                                                                                                                                                                                            | (20 marks) |
| (a) | <p>Define with examples</p> <ol style="list-style-type: none"> <li>Regular Graph</li> <li>Binary Tree</li> <li>Path</li> <li>Complete Graph</li> <li>Bipartite Graph</li> </ol>                                                                                                                                                                                                                            | CO6        |
| (b) | <p>Using Kruskal's algorithm, find a spanning tree with minimum weight from the graph below. Also calculate the total weight of spanning tree.</p>                                                                                                                                                                       |            |
| (c) | <p>Solve the recurrence relation by the method of generating functions</p> $a_{n+2} - 2a_{n+1} + a_n = 2^n, \text{ given } a_0 = 2 \text{ and } a_1 = 1$                                                                                                                                                                                                                                                   |            |